

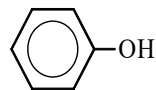
1 REVIEW FOR EXAM II Chem 1020

General things:

1. Study your notes; I have tried to organize the material putting emphasis on areas I believe are most important (and therefore will focus the exam on).
2. Read the chapters: emphasis on Chapter 13 and 14, but you are still responsible for 11 and 12.
3. Work the assigned homework problems. If you need repetition, turn your notebook to a clean page and *work the same problems again*. They should be easier and faster the second time around. If they aren't, you need to review more carefully.
4. You may expect an exam which has an objective section (multiple choice, fill in the blank), perhaps some short answer (one or two sentences), skills section (name, draw), complete the reaction (focus on the boxed sections at the end of the chapters) and an isomers question. Attention to spelling and grammar is expected.
5. You will **not** need a calculator. Pencil is preferred over pen. Model kits are permitted, although you should be aware that building the models often takes more time than you think.

Chapter 13 Alcohols, Phenols, Thiols, and Ethers

1. General Formula of Each : Alcohol: $R-OH$ Phenol:
(Thiols: $R-SH$) Ethers: $R-O-R'$



2. Properties: Solubility and m.p., b.p. of **ALL** of the major compounds in this unit compared to hydrocarbons. See Figure 14.2 on p. 465.
Presence of $-O-$ atoms makes cmpd sl. polar (little more water soluble), sl. higher m.p., b.p.,
Presence of $C=O$ group makes cmpds medium polar (mod water sol), med m.p., b.p.
Presence of $-OH$ groups makes cmpd very polar (very water soluble), much higher m.p., b.p.
3. Nomenclature: alcohols name ends in *-ol* and must tell where the $-OH$ group is;
ethers, *common* name: alkyl alkyl' ether (like methyl ethyl ether)
IUPAC name the smaller group as alkoxy, then alkane: 3-methoxyoctane
phenols are named like other benzene cmpds (e.g., 2-nitrophenol)
4. Specific Alcohols: you should know what is in the text and what I talked about:
methanol, ethanol, 2-propanol; toxicity and metabolism of alcohols
(a) ethanol is produced by fermentation of carbohydrates, or (b) hydration of ethylene (ethene)
5. Classification of alcohols: 1° (primary), 2° (secondary), 3° (tertiary)
6. Reactions: (1) combustion of alcohols - very clean burning
p. 450-51 (2) preparation of alcohols by hydration of alkenes
Rxn 1-5 (3a) dehydration of alcohols to form alkenes (reverse of 2) at $180^\circ C$
 (3b) dehydration of alcohols to form ethers at $140^\circ C$
 (4) oxidation of 1° alcohols to aldehydes and then carboxylic acids,
 2° alcohols to ketones;
 3° alcohols give N.R.
7. General oxid: hydrocarbons < alcohols, ethers < aldehydes, ketones < carboxylic acids, esters
8. Thiols - like an alcohol, but $-SH$ instead of $-OH$; malodorous, additive to natural gas, also called "mercaptans"; skunk oil



Chapter 14 - Aldehydes and Ketones the **C=O carbonyl group**

1. General Formula of aldehyde and ketone
2. Nomenclature : aldehydes end in *-al* and ketones end in *-one*.
(For ketones must tell by number where the C=O is, aldehydes understood as C-1)
3. Specific ones to know all about, e.g., formaldehyde (methanal) and acetone (propanone)
4. Oxidation of alcohols in biological processes (enzyme LAD in the liver)
ethanol $\rightarrow$ acetaldehyde $\rightarrow$ carbon dioxide Read in book about this.
methanol $\rightarrow$ formaldehyde (which is terribly toxic)
5. Reactions: **Oxidation** - 1° alcohols are easily oxidized to aldehydes, may be further oxidized to carboxylic acids (reaction often occurs so readily that it is difficult to *prevent*).
ketones do not oxidize; this is the basis of distinguishing aldehydes from ketones

Tollens' Test - production of silver mirror with aldehyde; with ketone, N.R.

Dichromate Test - reduction of Cr^{+6} (yellow soln) to Cr^{+3} (green soln)
1° alcohol (very fast), 2° alcohol (slow), 3° alcohol (N.R.);
aldehyde will react to give a carboxylic acid, ketone (N.R.)

Hydrogenation - addition of H_2 across the C=O (carbonyl) - this is a **reduction**
and the product is a 1° alcohol (from aldehyde) or 2° alcohol (from ketone)

Reactions 1-4 on p. 479-480 Chapter 14.